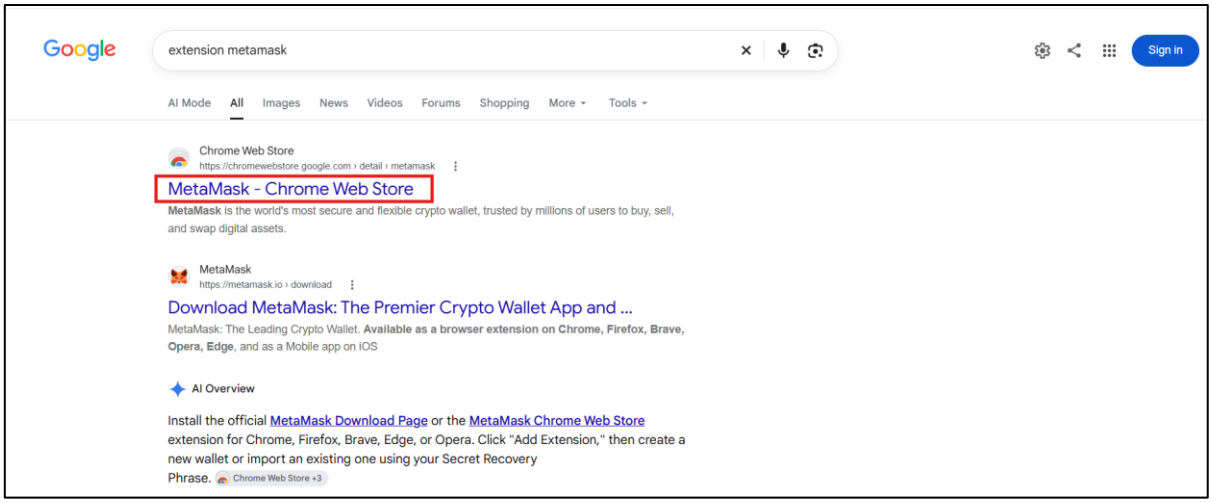
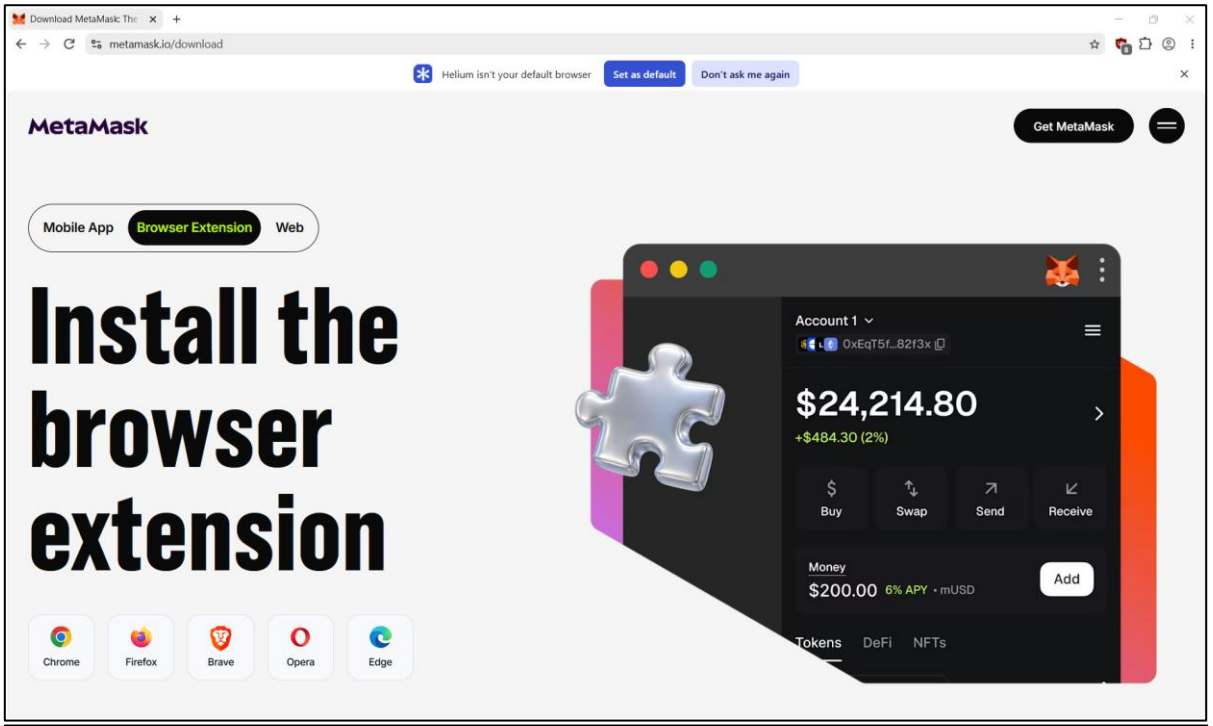
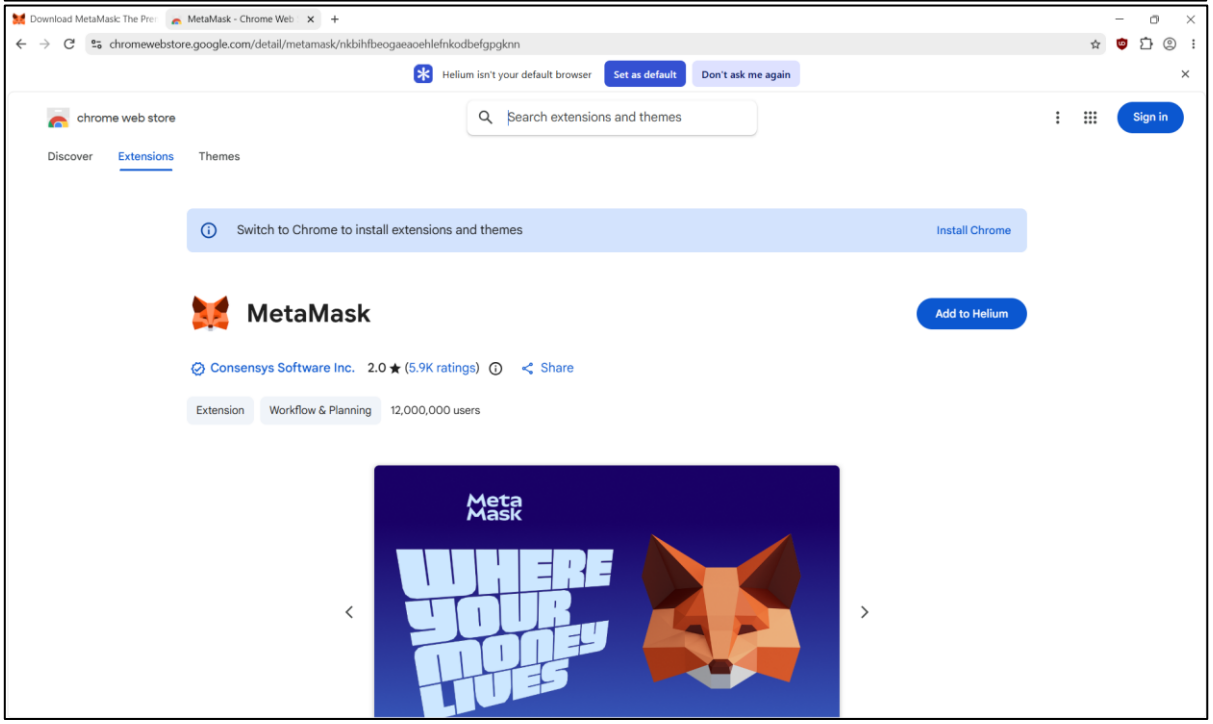
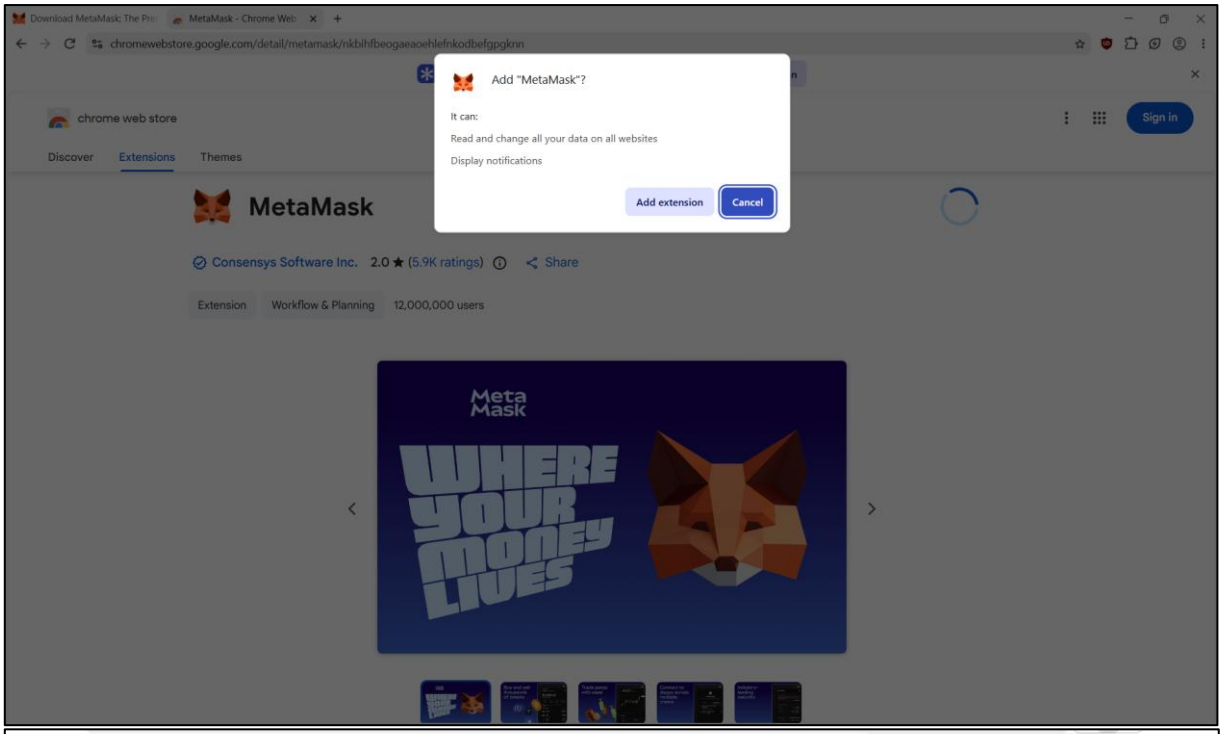
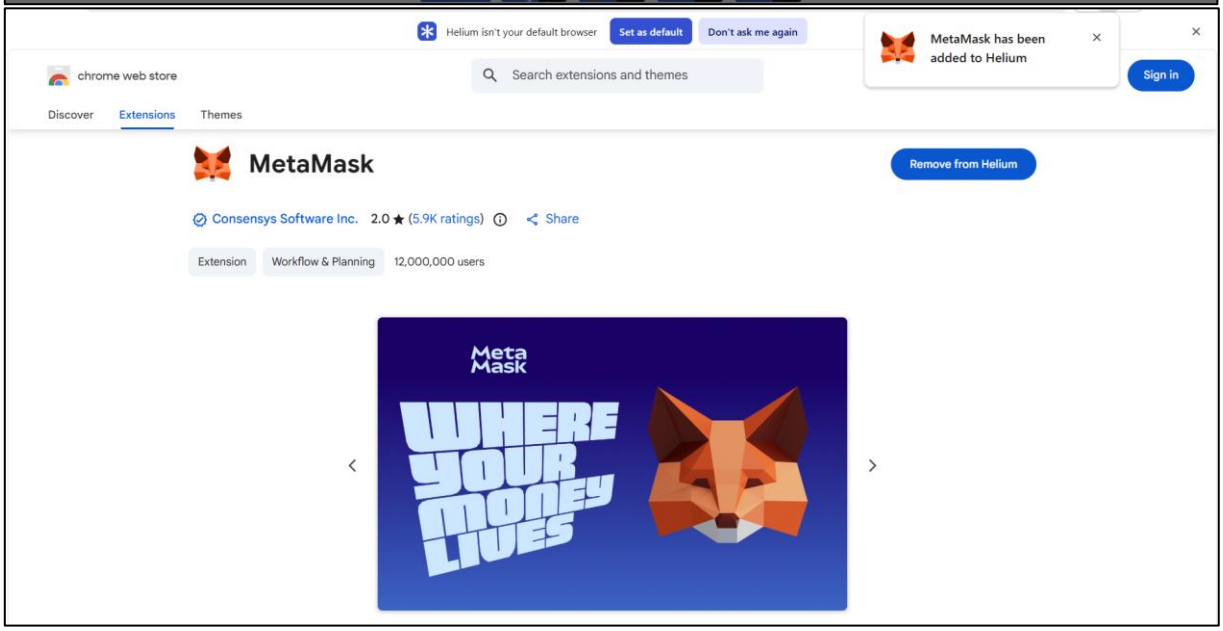
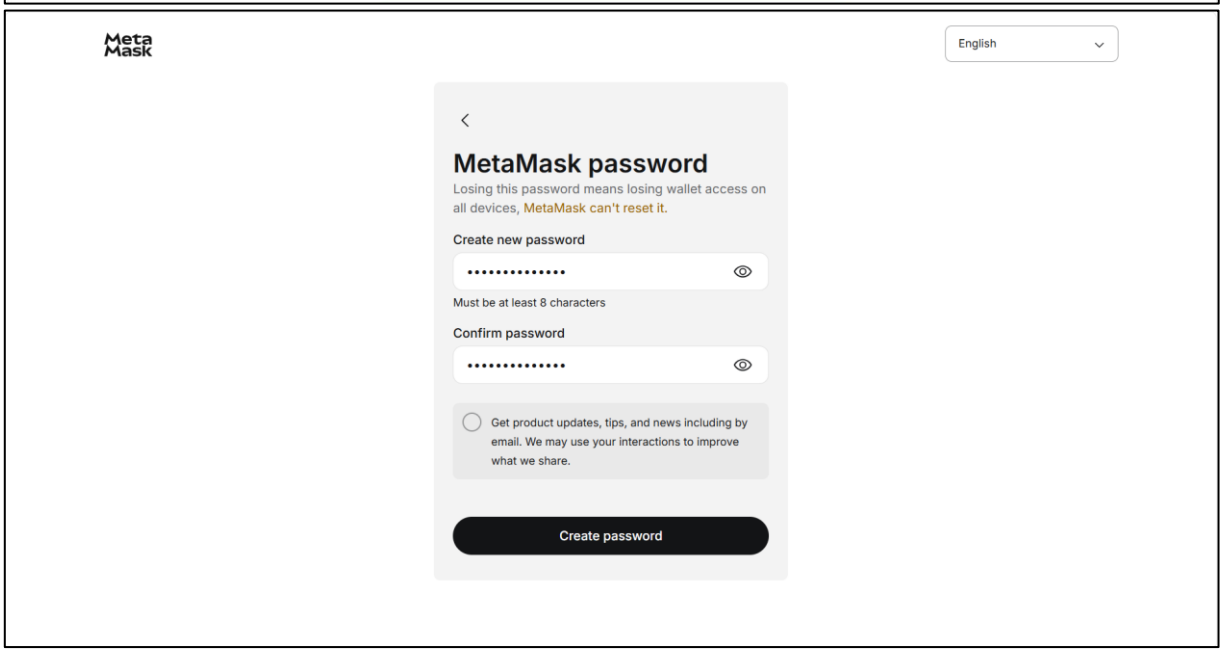

Google
extension metamask
Sign in
AI Mode
All
Images
News
Videos
Forums
Shopping
More
Tools
Chrome Web Store
https://chromewebstore.google.com › detail › metamask
MetaMask - Chrome Web Store
MetaMask is the world's most secure and flexible crypto wallet, trusted by millions of users to buy, sell, and swap digital assets.
MetaMask
https://metamask.io › download
Download MetaMask: The Premier Crypto Wallet App and ...
MetaMask: The Leading Crypto Wallet. Available as a browser extension on Chrome, Firefox, Brave, Opera, Edge, and as a Mobile app on iOS
AI Overview
Install the official MetaMask Download Page or the MetaMask Chrome Web Store extension for Chrome, Firefox, Brave, Edge, or Opera. Click "Add Extension," then create a new wallet or import an existing one using your Secret Recovery Phrase.
Chrome Web Store +3

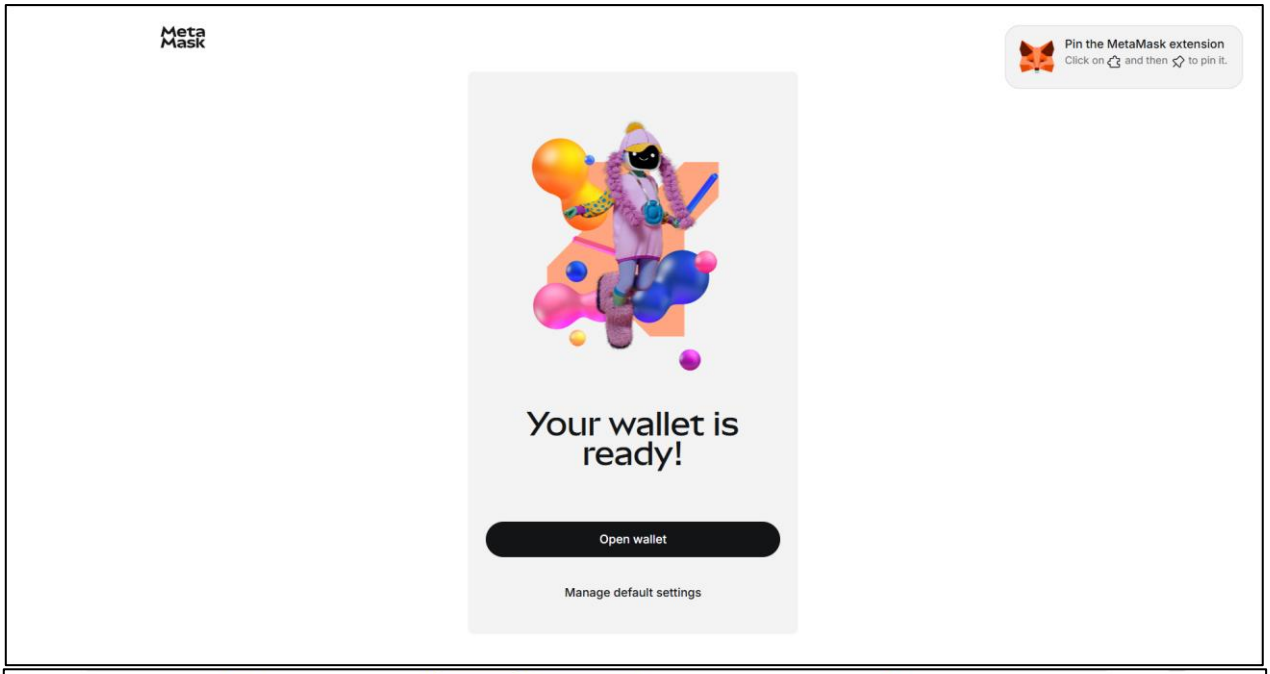

metamask.io/download
Helium isn't your default browser
Set as default
Don't ask me again
MetaMask
Get MetaMask
Mobile App
Browser Extension
Web
Install the browser extension
Chrome
Firefox
Brave
Opera
Edge
Account 1
0xEqT5f...82f3x
$24,214.80
+$484.30 (2%)
Buy
Swap
Send
Receive
Money
$200.00 6% APY • mUSD
Add
Tokens
DeFi
NFTs

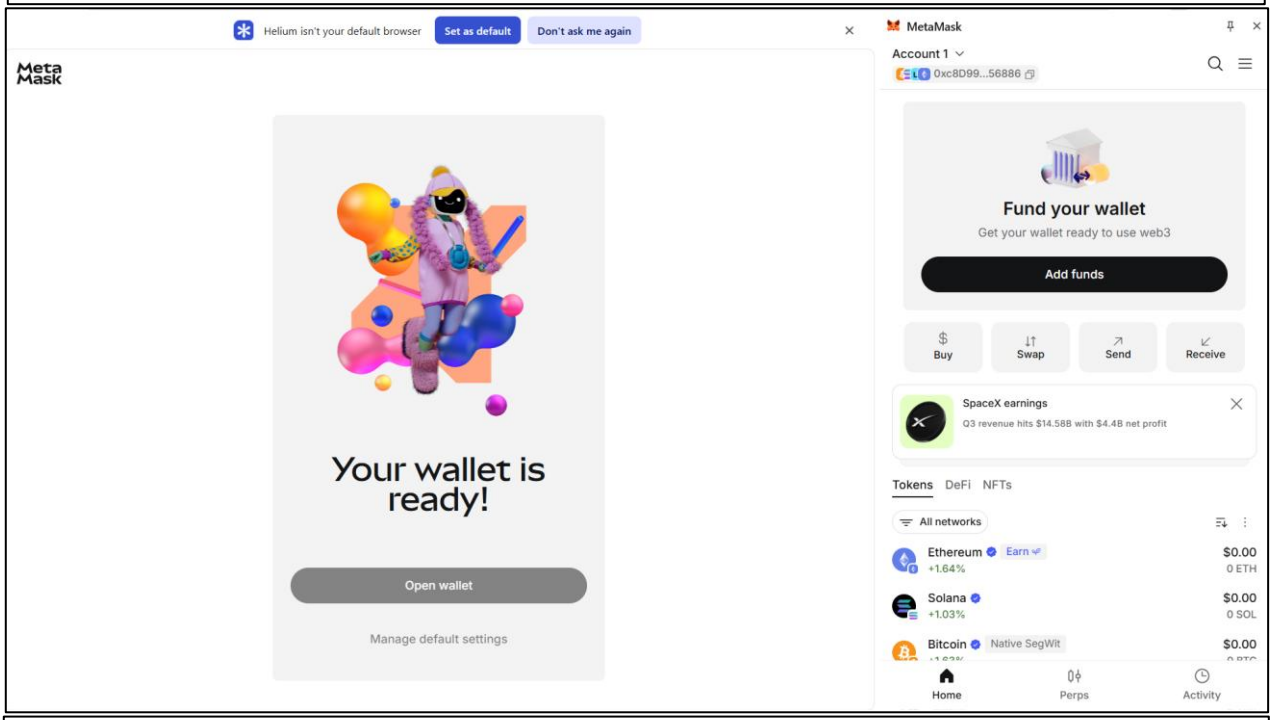

chromewebstore.google.com/detail/metamask/nkbihfbeogaeaoehlefnkodbefgpgknn
Helium isn't your default browser
Set as default
Don't ask me again
chrome web store
Search extensions and themes
Sign in
Discover
Extensions
Themes
Switch to Chrome to install extensions and themes
Install Chrome
MetaMask
Add to Helium
Consensys Software Inc.
2.0 ★ (5.9K ratings)
Share
Extension
Workflow & Planning
12,000,000 users
Meta Mask
WHERE YOUR MONEY LIVES

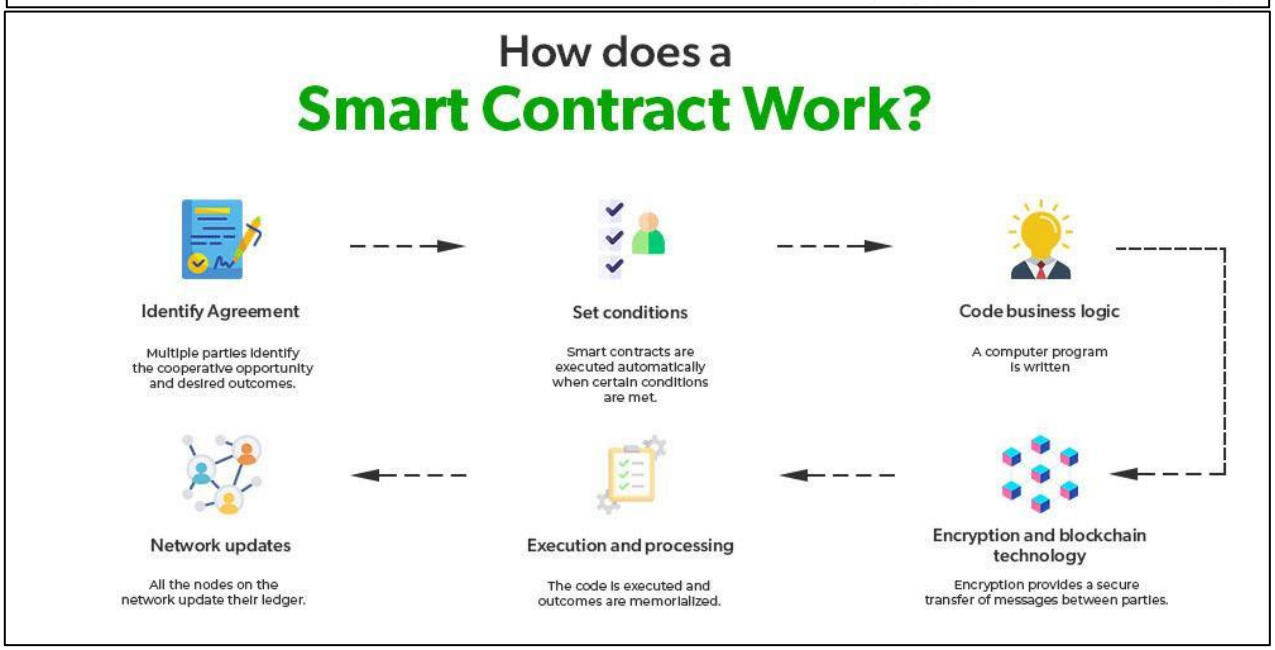

Add "MetaMask"?
It can:
Read and change all your data on all websites
Display notifications
Add extension
Cancel
chrome web store
Sign in
Discover
Extensions
Themes
MetaMask
Consensys Software Inc.
2.0 ★ (5.9K ratings)
Share
Extension
Workflow & Planning
12,000,000 users
Meta Mask
WHERE YOUR MONEY LIVES


Helium isn't your default browser
Set as default
Don't ask me again
MetaMask has been added to Helium
chrome web store
Search extensions and themes
Sign in
Discover
Extensions
Themes
MetaMask
Remove from Helium
Consensys Software Inc.
2.0 ★ (5.9K ratings)
Share
Extension
Workflow & Planning
12,000,000 users
Meta Mask
WHERE YOUR MONEY LIVES


Meta Mask
English
MetaMask password
Losing this password means losing wallet access on all devices, MetaMask can't reset it.
Create new password
Must be at least 8 characters
Confirm password
Get product updates, tips, and news including by email. We may use your interactions to improve what we share.
Create password

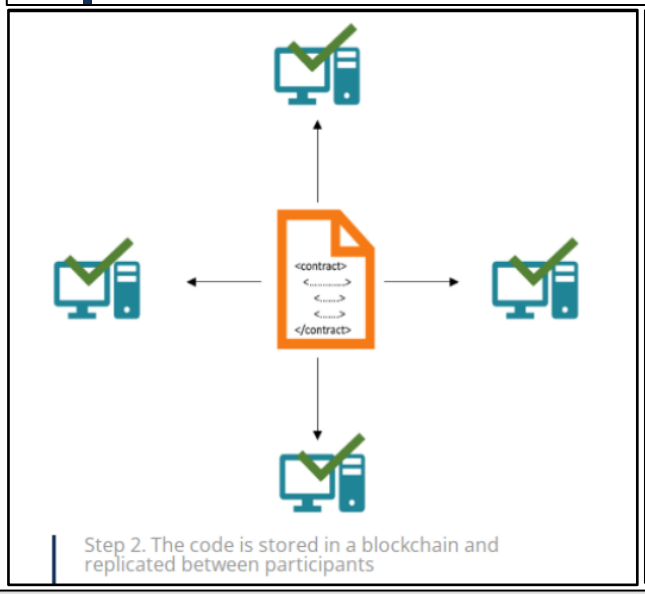

MetaMask
Pin the MetaMask extension
Click on and then to pin it.
Your wallet is ready!
Open wallet
Manage default settings


Helium isn't your default browser
Set as default
Don't ask me again
MetaMask
Your wallet is ready!
Open wallet
Manage default settings
MetaMask
Account 1
0xc8D99...56886
Fund your wallet
Get your wallet ready to use web3
Add funds
Buy
Swap
Send
Receive
SpaceX earnings
Q3 revenue hits $14.58B with $4.4B net profit
Tokens
DeFi
NFTs
All networks
Ethereum
Earn
+1.64%
$0.00
0 ETH
Solana
+1.03%
$0.00
0 SOL
Bitcoin
Native SegWit
$0.00
Home
Perps
Activity

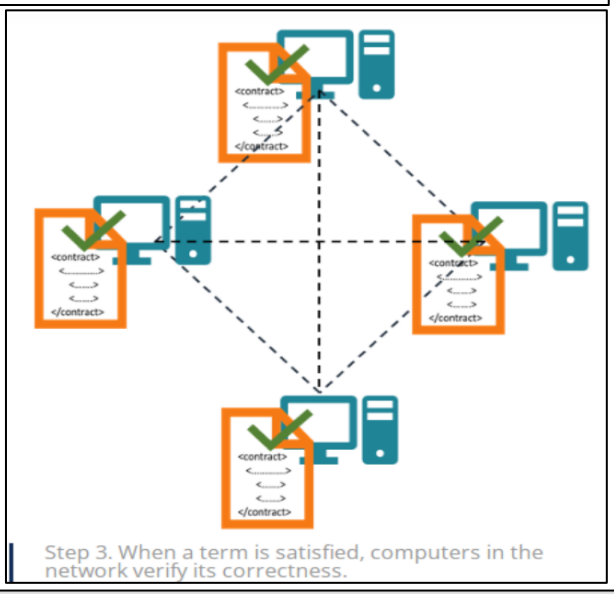

How does a
Smart Contract Work?
Identify Agreement
Multiple parties identify the cooperative opportunity and desired outcomes.
Set conditions
Smart contracts are executed automatically when certain conditions are met.
Code business logic
A computer program is written
Encryption and blockchain technology
Encryption provides a secure transfer of messages between parties.
Execution and processing
The code is executed and outcomes are memorialized.
Network updates
All the nodes on the network update their ledger.



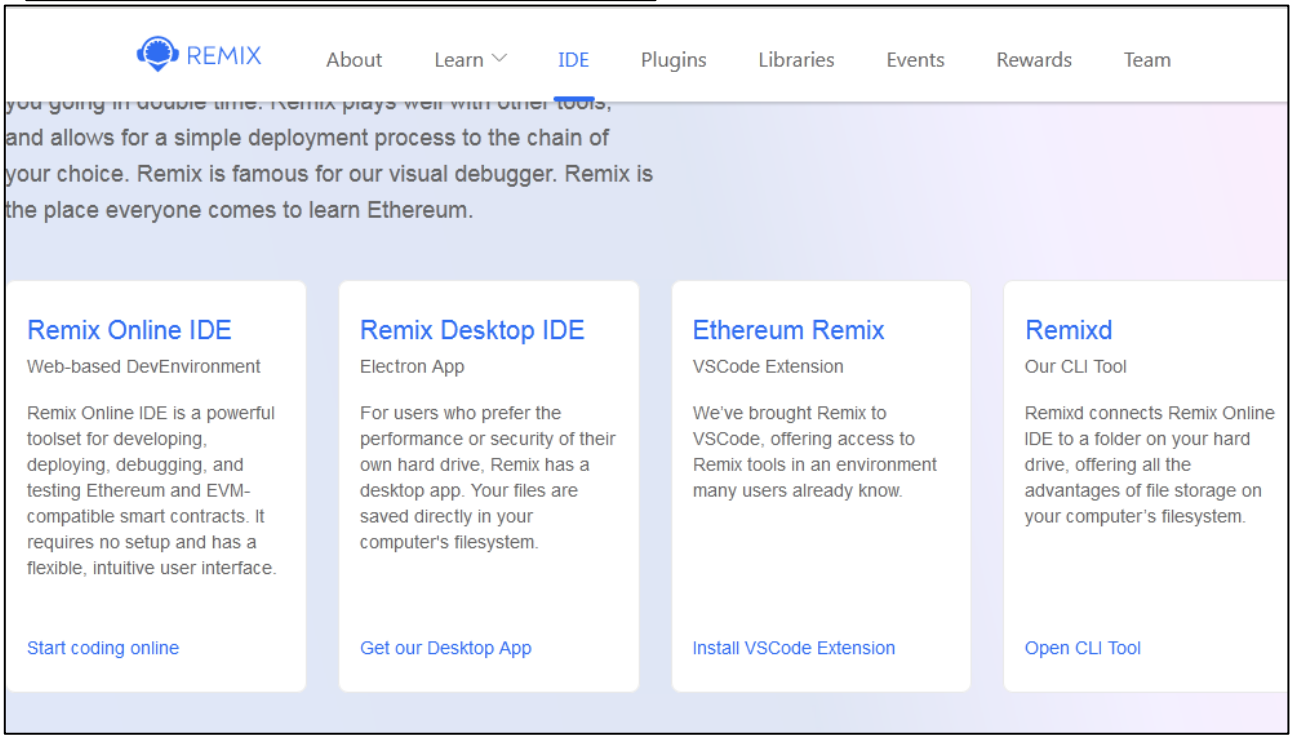
Step 1. Transfer contract terms into code

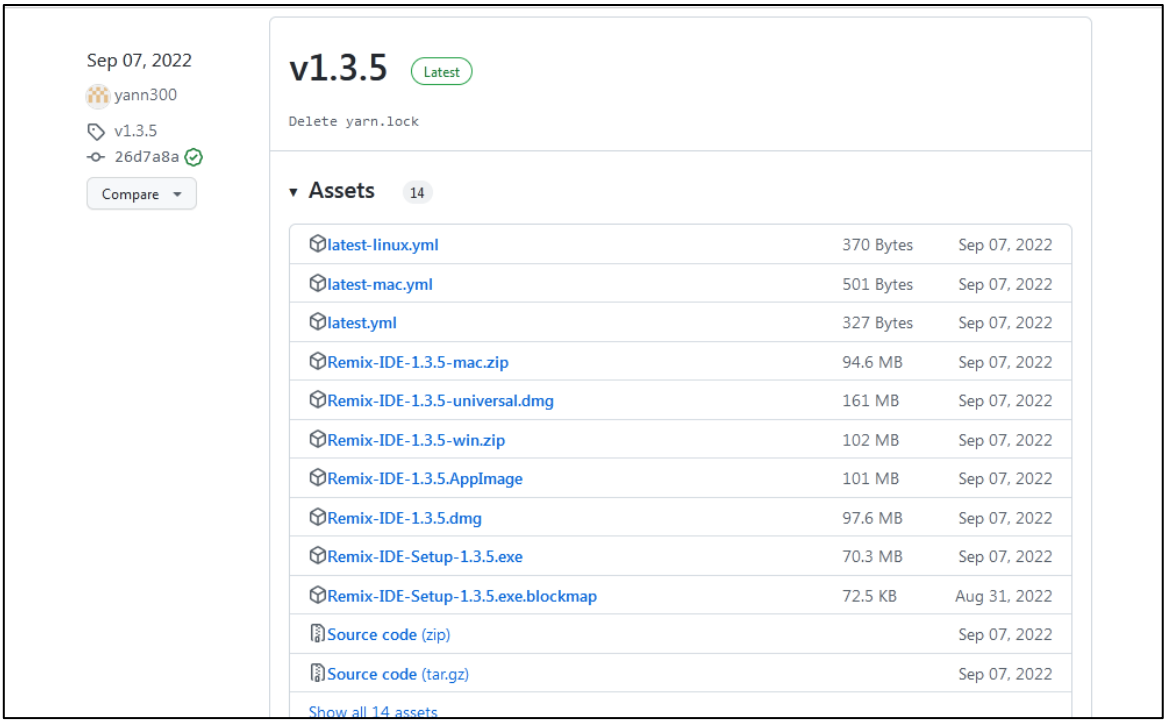


Step 2. The code is stored in a blockchain and replicated between participants

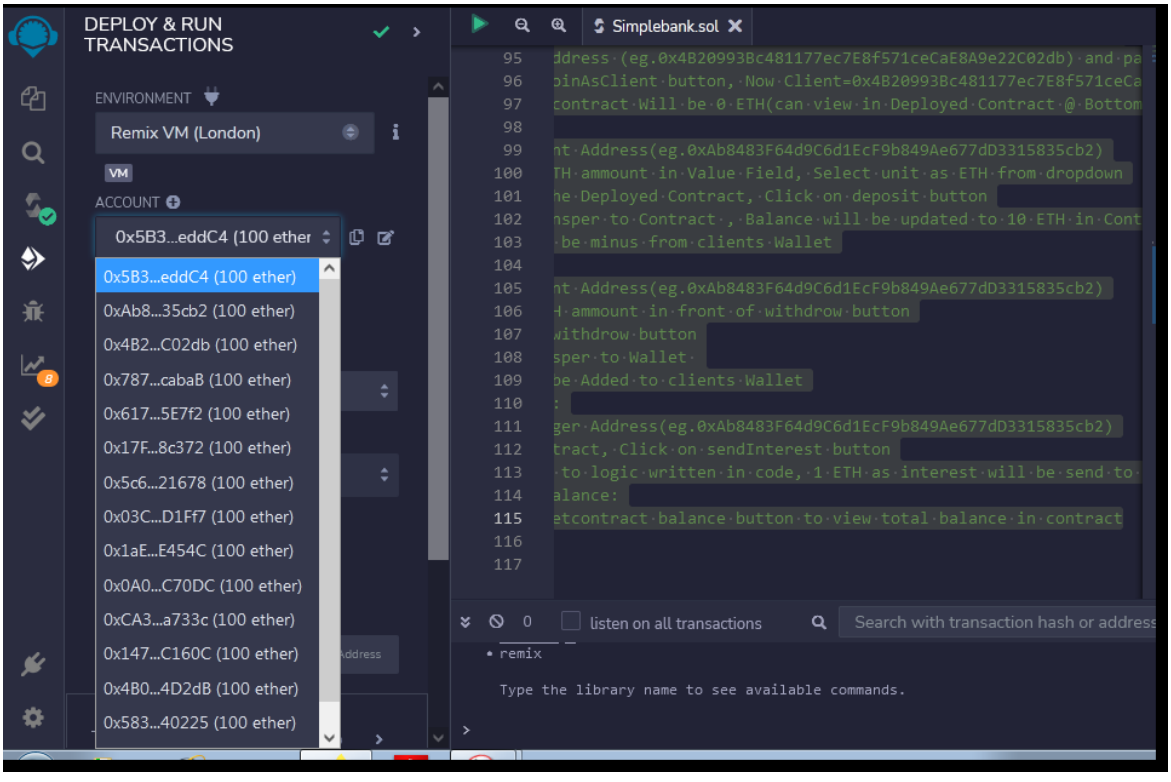


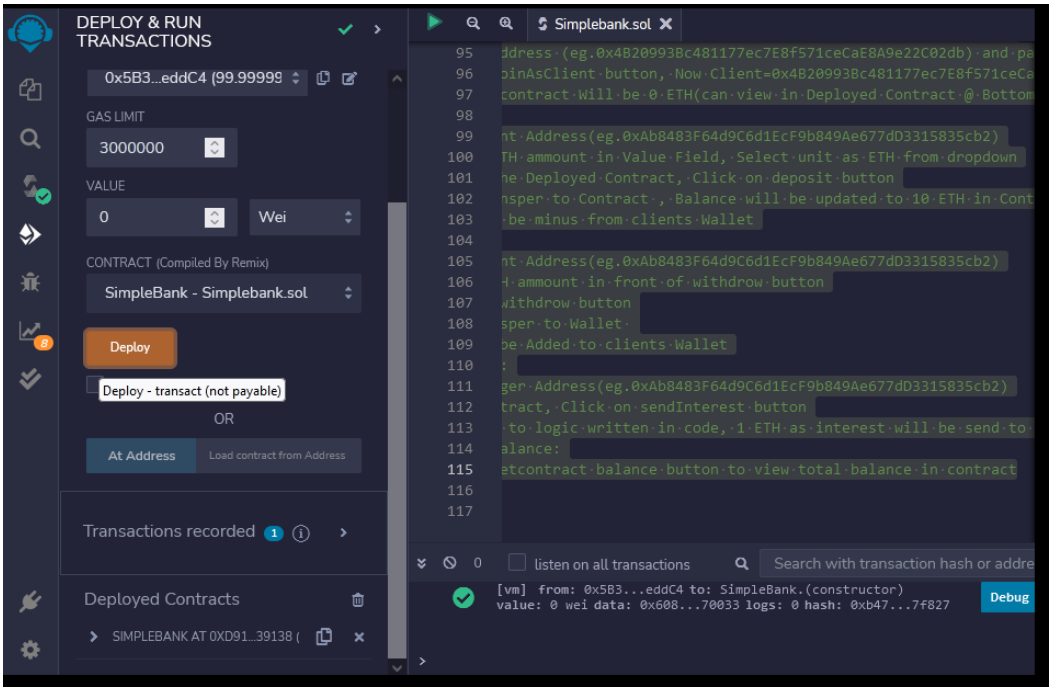
Step 3. When a term is satisfied, computers in the network verify its correctness.

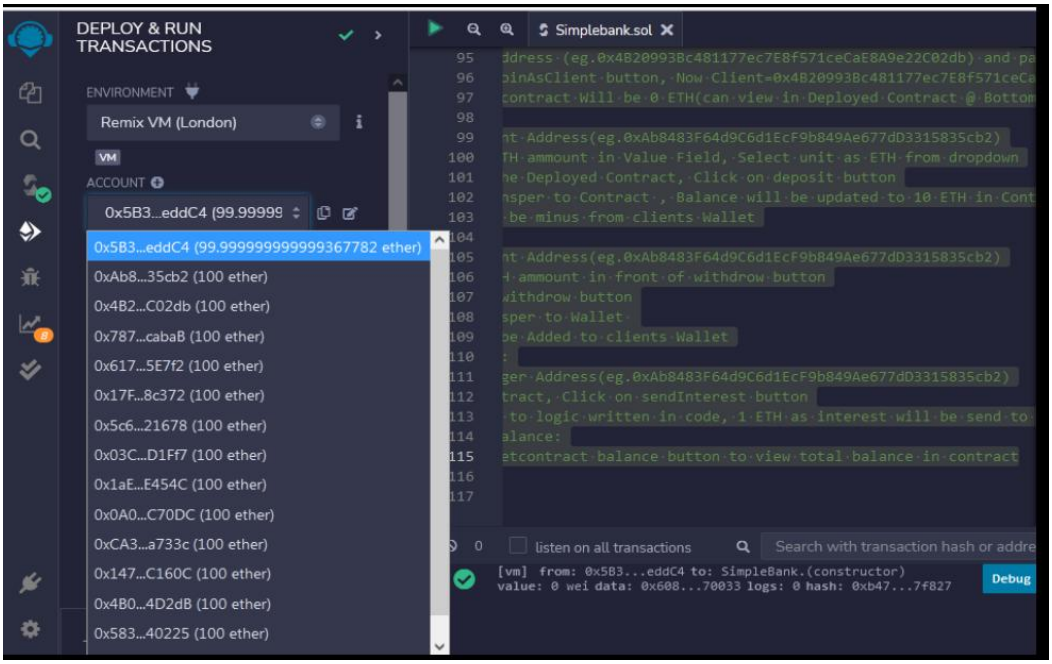


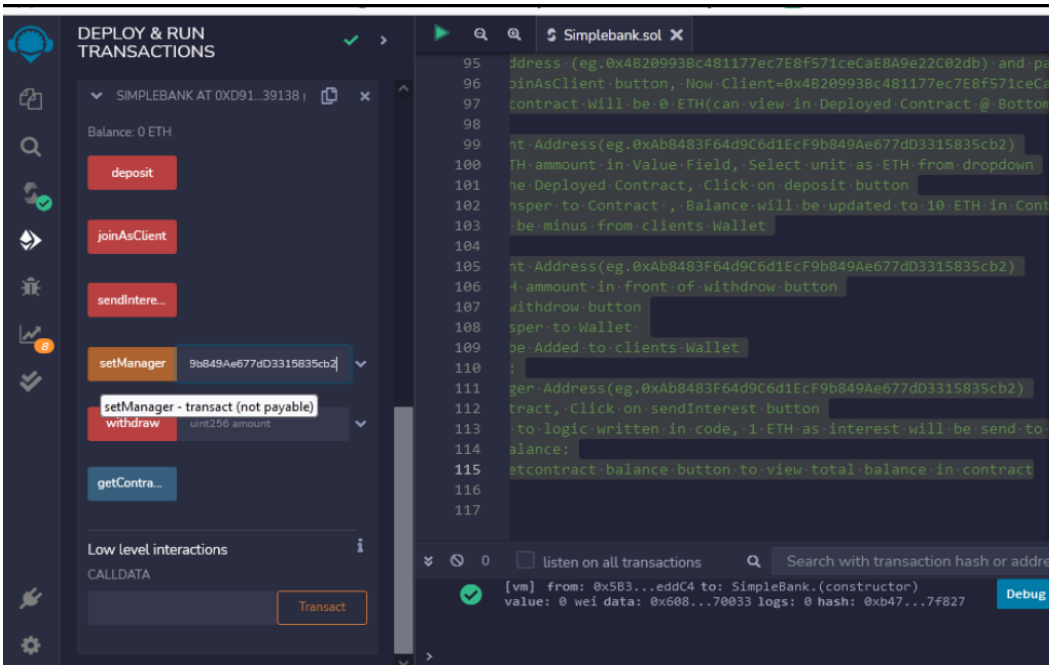


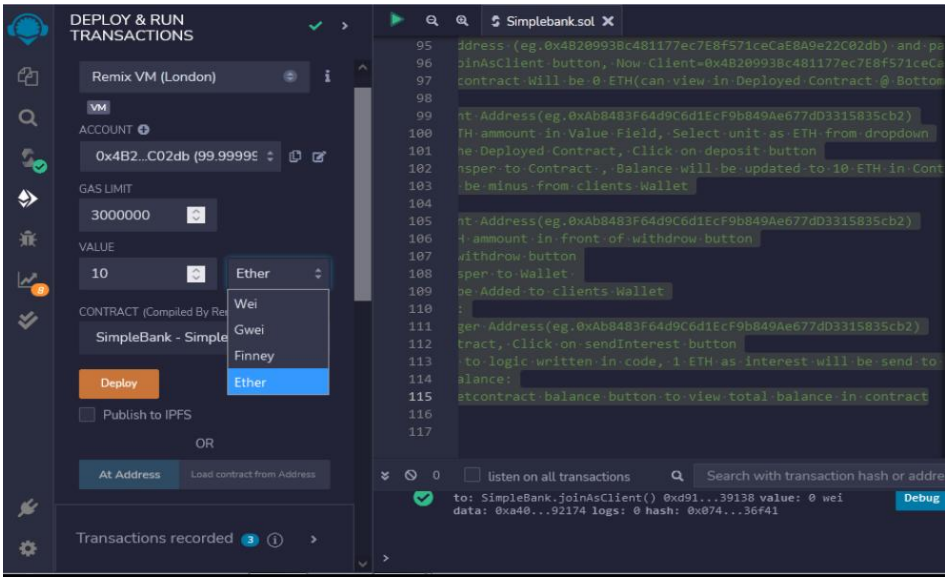

```
pragma solidity^0.4.0;
contract StorageBasic {
  uint storedValue;
  function set(uint var) {
     storedValue= var;
  }
  function get() constant returns (uint) {
   return storedValue;
  }
}
```

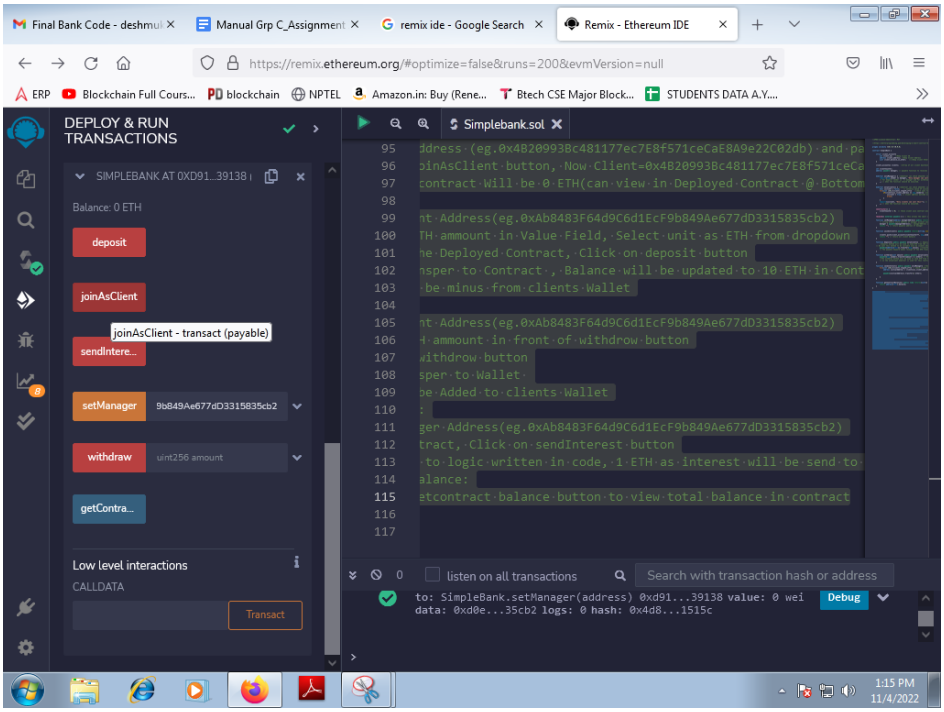


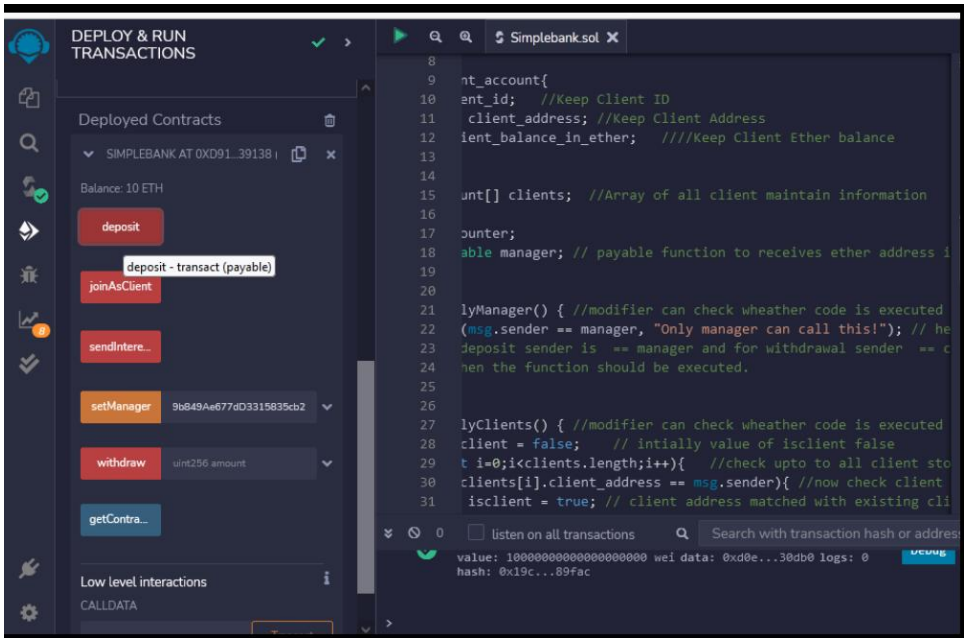

DEPLOY & RUN TRANSACTIONS
0x5B3...eddC4 (99.99999
GAS LIMIT
3000000
VALUE
0
Wei
CONTRACT (Compiled By Remix)
SimpleBank - Simplebank.sol
Deploy
Deploy - transact (not payable)
OR
At Address
Load contract from Address
Transactions recorded 1
Deployed Contracts
SIMPLEBANK AT 0XD91...39138 (
Simplebank.sol
listen on all transactions
Search with transaction hash or addre
[vm] from: 0x5B3...eddC4 to: SimpleBank.(constructor)
value: 0 wei data: 0x608...70033 logs: 0 hash: 0xb47...7f827
Debug

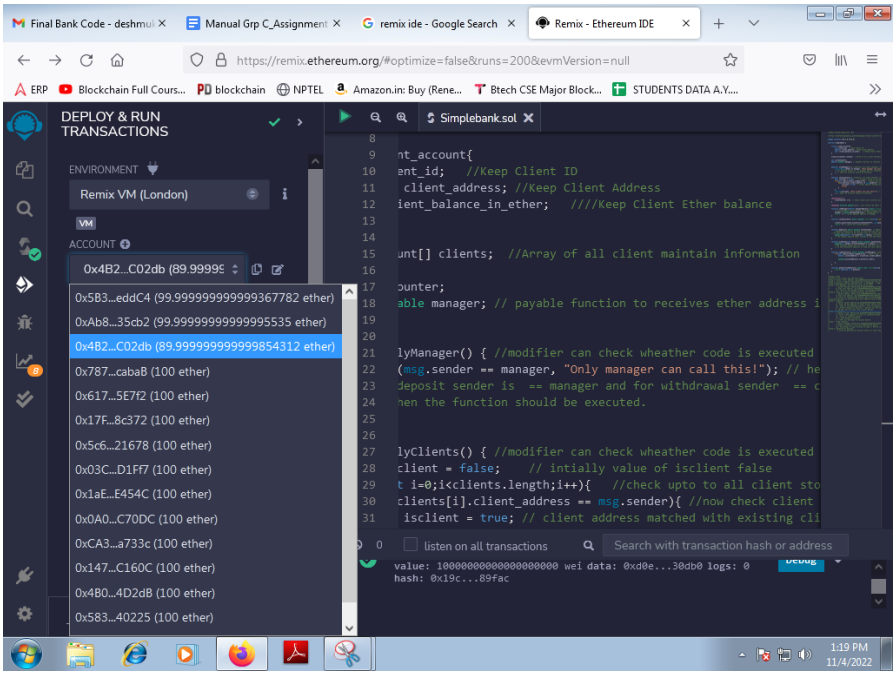

DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Remix VM (London)
VM
ACCOUNT
0x5B3...eddC4 (99.99999
0x5B3...eddC4 (99.999999999999367782 ether)
0xAb8...35cb2 (100 ether)
0x4B2...C02db (100 ether)
0x787...cabaB (100 ether)
0x617...5E7f2 (100 ether)
0x17F...8c372 (100 ether)
0x5c6...21678 (100 ether)
0x03C...D1Ff7 (100 ether)
0x1aE...E454C (100 ether)
0x0A0...C70DC (100 ether)
0xCA3...a733c (100 ether)
0x147...C160C (100 ether)
0x4B0...4D2dB (100 ether)
0x583...40225 (100 ether)
Simplebank.sol
listen on all transactions
Search with transaction hash or addre
[vm] from: 0x5B3...eddC4 to: SimpleBank.(constructor)
value: 0 wei data: 0x608...70033 logs: 0 hash: 0xb47...7f827
Debug

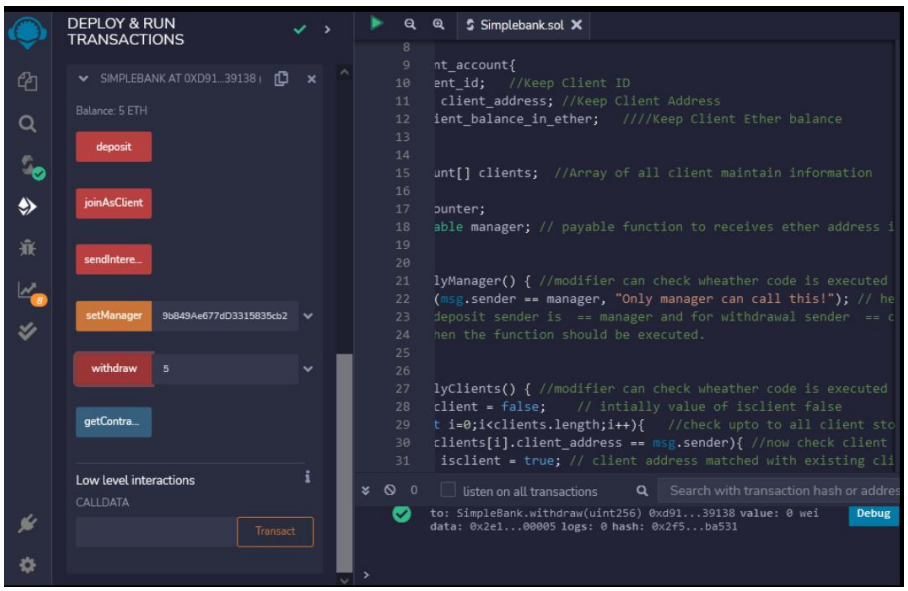

DEPLOY & RUN TRANSACTIONS
SIMPLEBANK AT 0XD91...39138 (
Balance: 0 ETH
deposit
joinAsClient
sendIntere...
setManager
9b849Ae677dD3315835cb2
setManager - transact (not payable)
withdraw
uint256 amount
getContra...
Low level interactions
CALLDATA
Transact
Simplebank.sol
listen on all transactions
Search with transaction hash or addre
[vm] from: 0x5B3...eddC4 to: SimpleBank.(constructor)
value: 0 wei data: 0x608...70033 logs: 0 hash: 0xb47...7f827
Debug

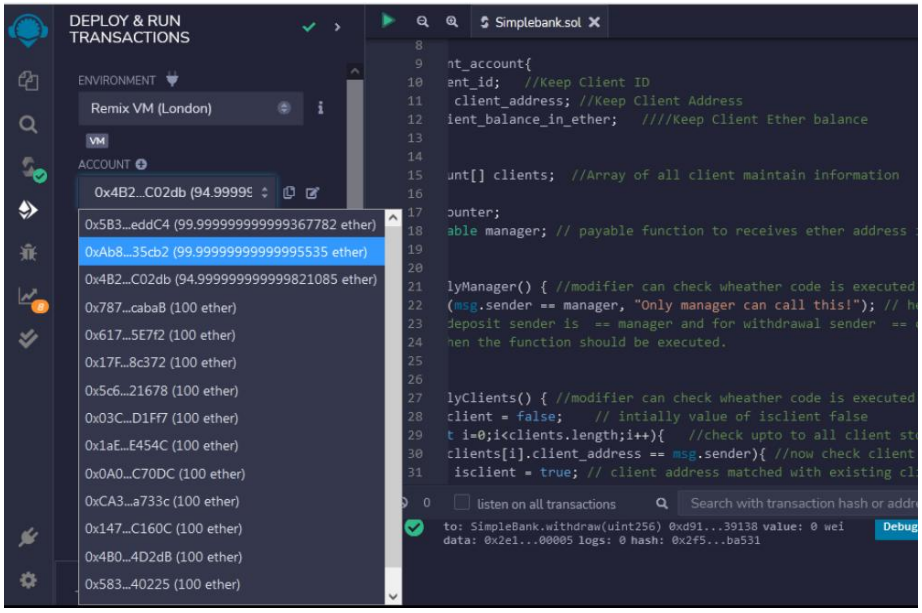

DEPLOY & RUN TRANSACTIONS
Remix VM (London)
VM
ACCOUNT
0x4B2...C02db (99.9999
GAS LIMIT
3000000
VALUE
10
Ether
Wei
Gwei
Finney
Ether
CONTRACT (Compiled By Re
SimpleBank - Simple
Deploy
Publish to IPFS
OR
At Address
Load contract from Address
Transactions recorded 3
Simplebank.sol
95 ddress (eg.0x4B20993Bc481177ec7E8f571ceCaE8A9e22C02db) and pa
96 oinAsClient button, Now Client=0x4B20993Bc481177ec7E8f571ceCa
97 contract Will be 0 ETH(can view in Deployed Contract @ Bottom
98
99 nt Address(eg.0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2)
100 TH ammount in Value Field, Select unit as ETH from dropdown
101 he Deployed Contract, Click on deposit button
102 nsper to Contract., Balance will be updated to 10 ETH in Cont
103 be minus from clients Wallet
104
105 nt Address(eg.0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2)
106 H ammount in front of withdrow button
107 withdrow button
108 sper to Wallet
109 be Added to clients Wallet
110 :
111 ger Address(eg.0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2)
112 tract, Click on sendInterest button
113 to logic written in code, 1 ETH as interest will be send to
114 alance:
115 etcontract balance button to view total balance in contract
116
117
listen on all transactions
Search with transaction hash or addre
to: SimpleBank.joinAsClient() 0xd91...39138 value: 0 wei
data: 0xa40...92174 logs: 0 hash: 0x074...36f41
Debug

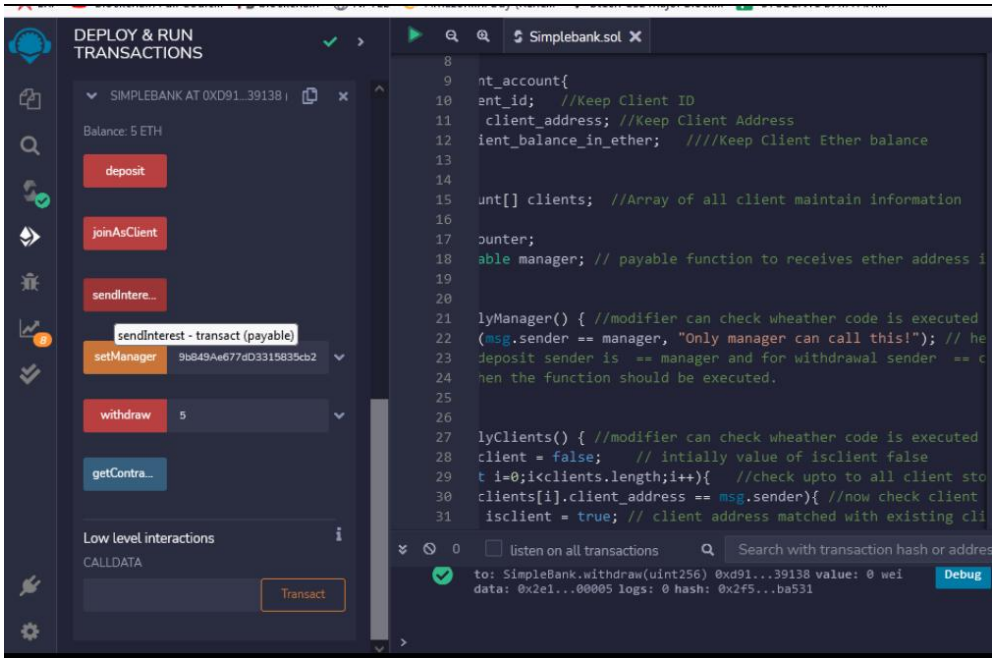

Final Bank Code - deshmul
Manual Grp C_Assignment
remix ide - Google Search
Remix - Ethereum IDE
https://remix.ethereum.org/#optimize=false&runs=200&evmVersion=null
ERP
Blockchain Full Cours...
blockchain
NPTEL
Amazon.in: Buy (Rene...
Btech CSE Major Block...
STUDENTS DATA A.Y....
DEPLOY & RUN TRANSACTIONS
SIMPLEBANK AT 0XD91...39138
Balance: 0 ETH
deposit
joinAsClient
joinAsClient - transact (payable)
sendIntere...
setManager
9b849Ae677dD3315835cb2
withdraw
uint256 amount
getContra...
Low level interactions
CALLDATA
Transact
Simplebank.sol
listen on all transactions
Search with transaction hash or address
to: SimpleBank.setManager(address) 0xd91...39138 value: 0 wei
data: 0xd0e...35cb2 logs: 0 hash: 0x4d8...1515c
Debug
1:15 PM
11/4/2022

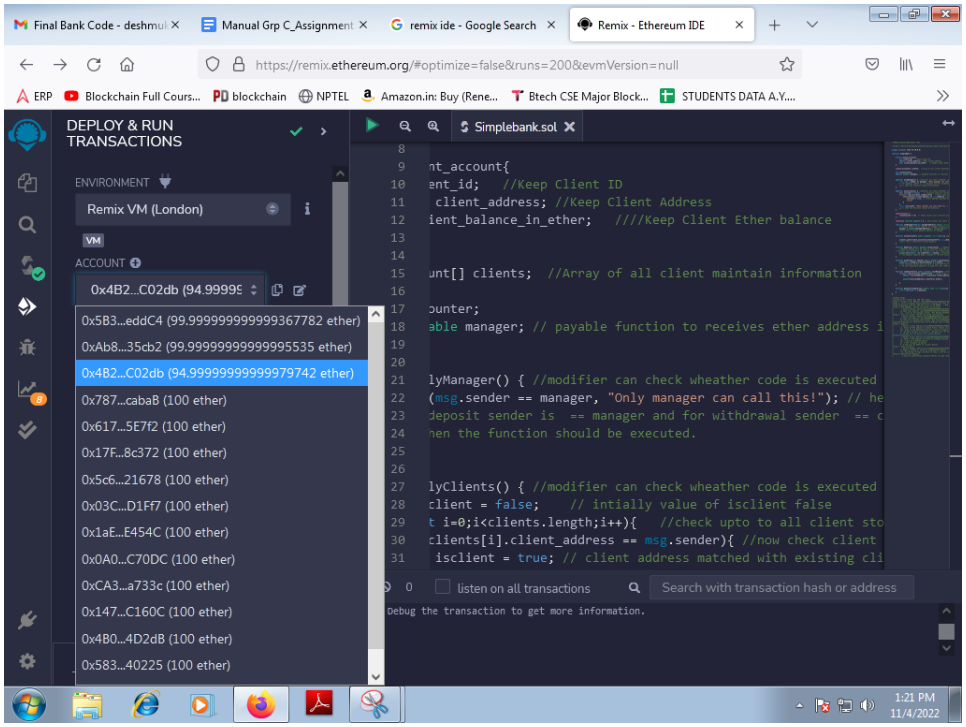

DEPLOY & RUN TRANSACTIONS
Deployed Contracts
SIMPLEBANK AT 0XD91...39138
Balance: 10 ETH
deposit
deposit - transact (payable)
joinAsClient
sendIntere...
setManager
9b849Ae677dD3315835cb2
withdraw
uint256 amount
getContra...
Low level interactions
CALLDATA
Simplebank.sol
nt_account{
ent_id; //Keep Client ID
client_address; //Keep Client Address
ient_balance_in_ether; ////Keep Client Ether balance
unt[] clients; //Array of all client maintain information
ounter;
able manager; // payable function to receives ether address i
lyManager() { //modifier can check wheather code is executed
(msg.sender == manager, "Only manager can call this!"); // he
deposit sender is == manager and for withdrawal sender ==
hen the function should be executed.
lyClients() { //modifier can check wheather code is executed
client = false; // intially value of isclient false
t i=0;i<clients.length;i++){ //check upto to all client sto
clients[i].client_address == msg.sender){ //now check client
isclient = true; // client address matched with existing cli
listen on all transactions
Search with transaction hash or addres
value: 1000000000000000000 wei data: 0xd0e...30db0 logs: 0
hash: 0x19c...89fac

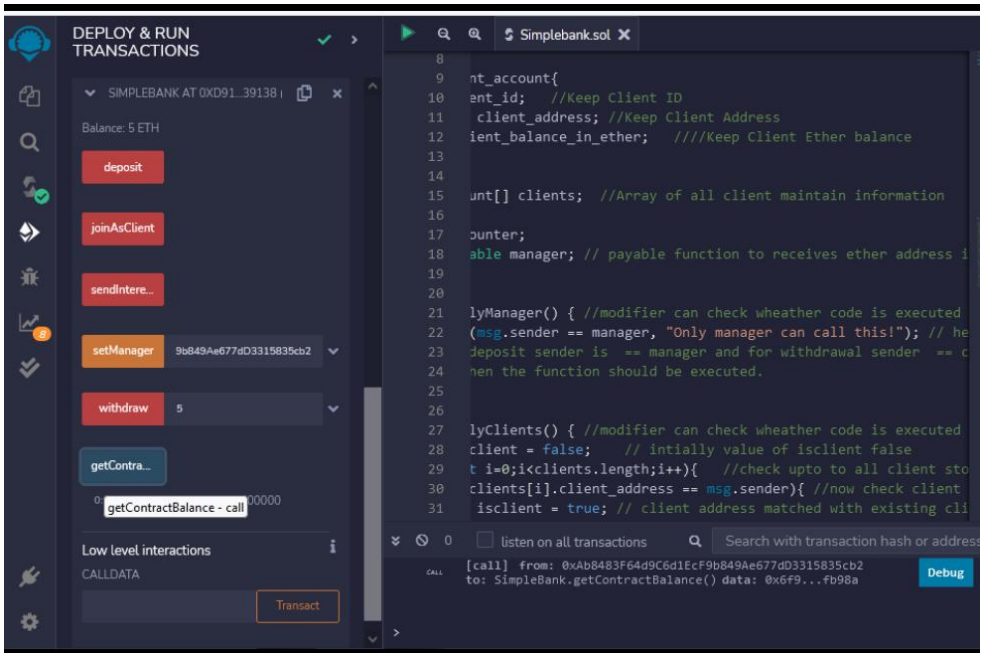

DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Remix VM (London)
VM
ACCOUNT
0x4B2...C02db (89.9999
0x5B3...eddC4 (99.99999999999367782 ether)
0xAb8...35cb2 (99.99999999999995535 ether)
0x4B2...C02db (89.999999999999854312 ether)
0x787...cabaB (100 ether)
0x617...5E7f2 (100 ether)
0x17F...8c372 (100 ether)
0x5c6...21678 (100 ether)
0x03C...D1Ff7 (100 ether)
0x1aE...E454C (100 ether)
0x0A0...C70DC (100 ether)
0xCA3...a733c (100 ether)
0x147...C160C (100 ether)
0x4B0...4D2dB (100 ether)
0x583...40225 (100 ether)
Simplebank.sol
listen on all transactions
Search with transaction hash or address
1:19 PM
11/4/2022


DEPLOY & RUN TRANSACTIONS
SIMPLEBANK AT 0XD91...39138
Balance: 5 ETH
deposit
joinAsClient
sendIntere...
setManager
9b849Ae677dD3315835cb2
withdraw
5
getContra...
Low level interactions
CALLDATA
Transact
Simplebank.sol
listen on all transactions
Search with transaction hash or addres
to: SimpleBank.withdraw(uint256) 0xd91...39138 value: 0 wei
data: 0x2e1...00005 logs: 0 hash: 0x2f5...ba531
Debug


DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Remix VM (London)
VM
ACCOUNT
0x4B2...C02db (94.9999
0x5B3...eddC4 (99.99999999999367782 ether)
0xAb8...35cb2 (99.99999999999995535 ether)
0x4B2...C02db (94.999999999999821085 ether)
0x787...cabaB (100 ether)
0x617...5E7f2 (100 ether)
0x17F...8c372 (100 ether)
0x5c6...21678 (100 ether)
0x03C...D1Ff7 (100 ether)
0x1aE...E454C (100 ether)
0x0A0...C70DC (100 ether)
0xCA3...a733c (100 ether)
0x147...C160C (100 ether)
0x4B0...4D2dB (100 ether)
0x583...40225 (100 ether)
Simplebank.sol
listen on all transactions
Search with transaction hash or addr
to: SimpleBank.withdraw(uint256) 0xd91...39138 value: 0 wei
data: 0x2e1...00005 logs: 0 hash: 0x2f5...ba531
Debug

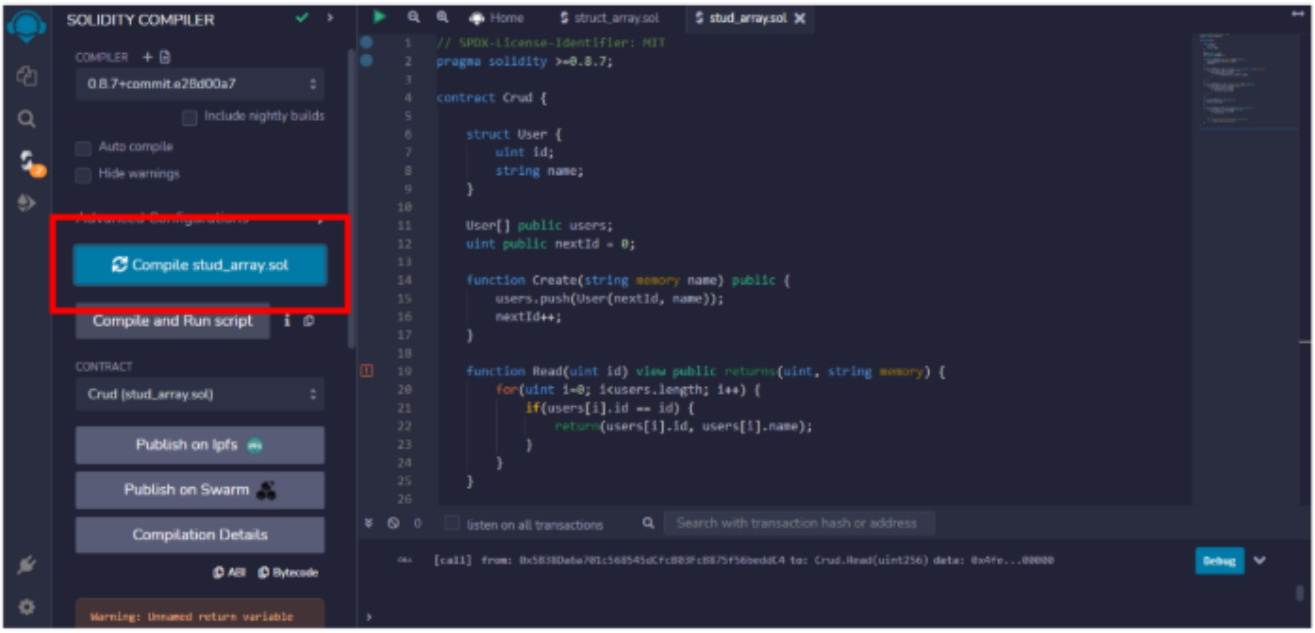

DEPLOY & RUN TRANSACTIONS
SIMPLEBANK AT 0XD91...39138
Balance: 5 ETH
deposit
joinAsClient
sendIntere...
sendInterest - transact (payable)
setManager
9b849Ae677dD3315835cb2
withdraw
5
getContra...
Low level interactions
CALLDATA
Transact
Simplebank.sol
listen on all transactions
Search with transaction hash or addres
to: SimpleBank.withdraw(uint256) 0xd91...39138 value: 0 wei
data: 0x2e1...00005 logs: 0 hash: 0x2f5...ba531
Debug


Final Bank Code - deshmu
Manual Grp C_Assignment
remix ide - Google Search
Remix - Ethereum IDE
https://remix.ethereum.org/#optimize=false&runs=200&evmVersion=null
DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Remix VM (London)
VM
ACCOUNT
0x4B2...C02db (94.99999
0x5B3...eddC4 (99.999999999999367782 ether)
0xAb8...35cb2 (99.99999999999995535 ether)
0x4B2...C02db (94.99999999999979742 ether)
0x787...cabaB (100 ether)
0x617...5E7f2 (100 ether)
0x17F...8c372 (100 ether)
0x5c6...21678 (100 ether)
0x03C...D1Ff7 (100 ether)
0x1aE...E454C (100 ether)
0x0A0...C70DC (100 ether)
0xCA3...a733c (100 ether)
0x147...C160C (100 ether)
0x4B0...4D2dB (100 ether)
0x583...40225 (100 ether)
Simplebank.sol
listen on all transactions
Search with transaction hash or address
Debug the transaction to get more information.
1:21 PM
11/4/2022

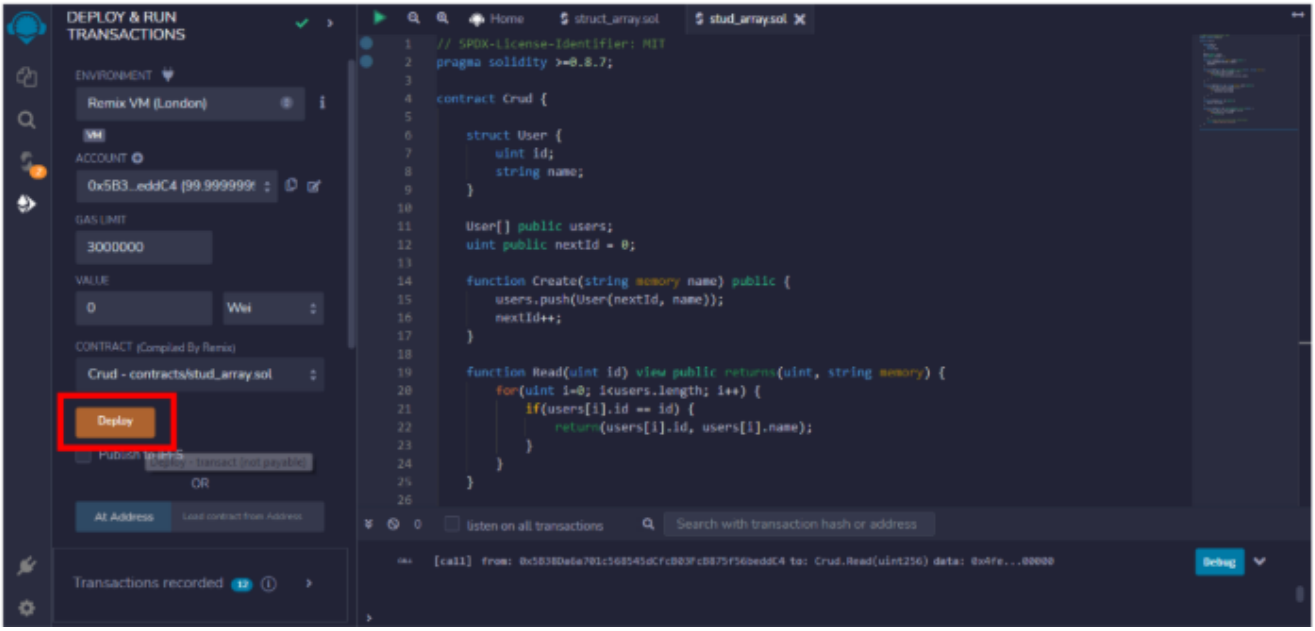

DEPLOY & RUN TRANSACTIONS
SIMPLEBANK AT 0XD91...39138
Balance: 5 ETH
deposit
joinAsClient
sendIntere...
setManager
9b849Ae677dD3315835cb2
withdraw
5
getContra...
getContractBalance - call
Low level interactions
CALLDATA
Transact
Simplebank.sol
listen on all transactions
Search with transaction hash or address
[call] from: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2
to: SimpleBank.getContractBalance() data: 0x6f9...fb98a
Debug

```
pragma solidity ^0.5.0;
contract Crud {
    struct User {
        uint id;
        string name;
    }
    User[] public users;
    uint public nextId = 0;
    function Create(string memory name) public {
        users.push(User(nextId, name));
        nextId++;
    }
    function Read(uint id) view public returns(uint, string memory) {
        for(uint i=0; i<users.length; i++) {
            if(users[i].id == id) {
                return(users[i].id, users[i].name);
            }
        }
    }
    function Update(uint id, string memory name) public {
        for(uint i=0; i<users.length; i++) {
            if(users[i].id == id) {
                users[i].name =name;
            }
        }
    }
    function Delete(uint id) public {
        delete users[id];
    }
    function find(uint id) view internal returns(uint) {
        for(uint i=0; i< users.length; i++) {
            if(users[i].id == id) {
                return i;
            }
        }
        // if user does not exist then revert back
        revert("User does not exist");
    }
}
```

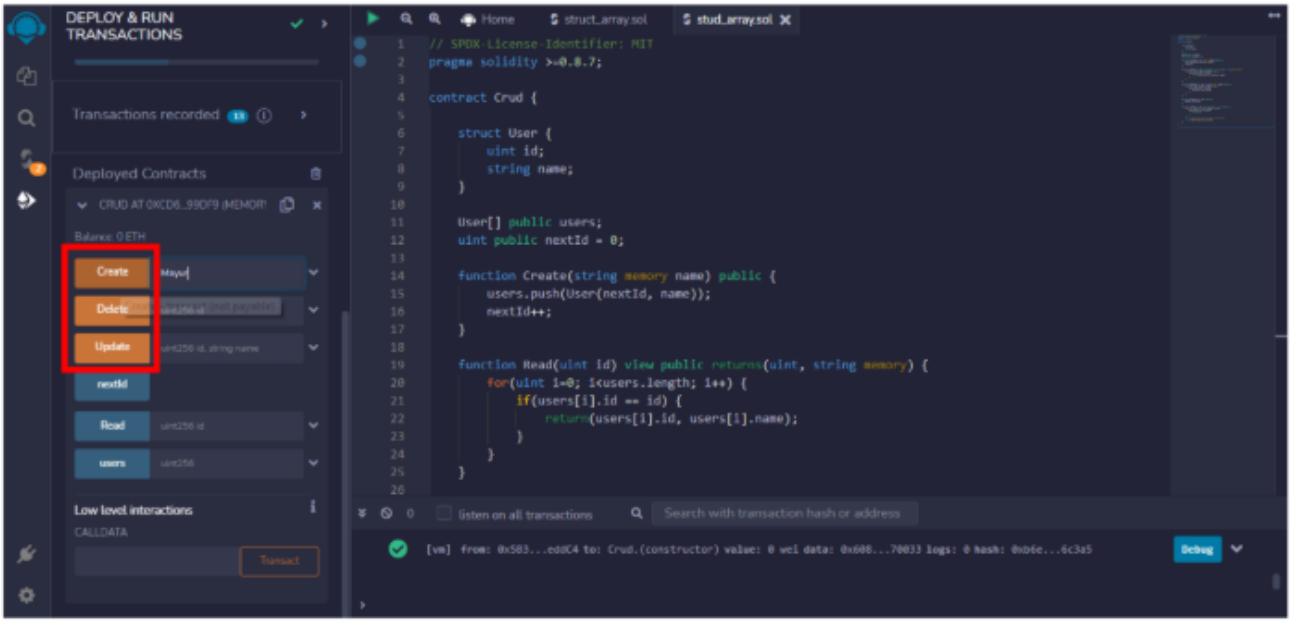
Step 1 – Compile the program by clicking on compile button.



Step 2 – After the Successful compilation, Deploy the contract to see the output.



Step 3 - Now you can check the output. You can insert, update and delete the student data using your smart contract.



Step 4 – After entering your data, you can read the data using ID.

